CareFlow — Week 4 Progress Report

Project: Clinical Pathway Process Mining **Focus:** Conformance checking and dashboard drill-down

1. Objective

Week 4 prepares a transparent first-pass comparison between each recorded patient journey and a defined reference pathway. Deviations are flagged at case level, and summaries are prepared for dashboard exploration.

2. Reference pathway

```
Registration → Triage → Physician Assessment → X-Ray → Physician Reassessment → Treatment → Discharge
```

A case conforms only when its timestamp-ordered activity sequence exactly matches this list. Extra activities, missing activities, and different ordering are flagged. This is a prototype exact-sequence rule, not formal alignment-based conformance and not a clinically approved standard.

3. Work prepared

• Required-column, blank-value, and timestamp validation.

• Timestamp-ordered traces grouped by Case_ID.

• Case-level conformance status, deviation flag, observed trace, explanation, and case duration.

• Overall conformance/deviation rates and mean case duration.

• Optional summaries by Department, Resource_ID, Age_Band, and Arrival_Shift when those columns exist.

• CSV exports for case results and cohort summaries, plus a JSON KPI summary.

• Automated tests for conforming traces, extra/missing activities, invalid timestamps, and missing drill-down fields.

4. Deliverables

File	Purpose
week4_conformance.py	Reusable conformance analysis and exports
tests/test_week4_conformance.py	Automated Week 4 tests
WEEK4_DAILY_COMMITS.md	Seven-day task and commit plan
WEEK4_PROGRESS_REPORT.md	Full written progress report

5. Test and sample-run status

The dedicated Week 4 test suite passed 6 tests in the development environment. Running the included synthetic event log produced 500 cases: 300 exact baseline matches (60.00%) and 200 deviations (40.00%). These are results from the included synthetic demonstration data only, not real hospital performance evidence.

6. Remaining integration work

A live BigQuery/dbt pipeline and published Power BI/Looker dashboard are not claimed as configured or verified. CSV/JSON outputs are prepared for downstream use, but dashboard filters and visuals, cloud execution, and KPI reconciliation must be implemented and verified in the target environment. Formal PM4Py alignment-based conformance can be considered after the model and approved pathway variants are validated.

7. Limitations and responsible use

The demonstration data is synthetic. The strict baseline can flag legitimate clinical alternatives as deviations. Before interpreting deviation rates as quality measures, obtain authorization, de-identify records, and agree valid pathway variants with clinical process owners. Never commit real patient data, credentials, or secrets to GitHub.

8. Next steps

Review pathway variants with process owners; run only on authorized, de-identified data; verify exported KPIs against dashboard calculations; configure drill-downs and cloud integrations; and document assumptions for any formal

conformance method.